

BEC503 Digital Communication — Syllabus mapping

EC branch, semester 5, 2025 scheme. Prescribed textbook: Simon Haykin, *Digital Communication Systems*, John Wiley & Sons, 1st edition, 2014.

Course outcomes

CO	Statement
CO1	Apply the concept of signal conversion to vectors in communication transmission and reception.
CO2	Perform the mathematical analysis of digital communication systems for different modulation techniques.
CO3	Apply the source coding and channel coding principles for the discrete memoryless channels.
CO4	Compute the codewords for the error correction and detection of a digital data using Linear Block Code, Cyclic Codes and Convolution Codes.
CO5	Design encoding and decoding circuits for Linear Block Code, Cyclic Codes and Convolution Codes.

Module plan

Module	Title	Hours	Chapters	Units	Deck slides	COs
1	Bandpass Signals to Equivalent Lowpass and Signalling over AWGN Channels	10	TB1 (Haykin 2014) Ch. 2.8-2.11 pp. 42-52 and Ch. 7.1-7.4 pp. 323-344	16	56	CO1
2	Digital Modulation Techniques	10	TB1 (Haykin 2014) Ch. 7.5-7.8 pp. 344-398 and Ch. 7.10-7.14 pp. 400-418	16	56	CO2
3	Information Theory	10	TB1 (Haykin 2014) Ch. 5.1-5.10 pp. 207-244	16	56	CO3
4	Error Control Coding: Linear Block and Cyclic Codes	10	TB1 (Haykin 2014) Ch. 10.1-10.5 pp. 577-605	16	56	CO4, CO5
5	Convolutional Codes	10	TB1 (Haykin 2014) Ch. 10.6-10.8 pp. 605-623	16	56	CO4, CO5

Module 1 — Bandpass Signals to Equivalent Lowpass and Signalling over AWGN Channels

Driving question: How do I turn a radio-frequency waveform into a handful of numbers, and what is the best possible receiver once I have?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Why a bandpass signal needs a lowpass equivalent	Ch. 2	42-45	5, 6, 7	CO1	L2
2	The Hilbert transform as a wideband 90-degree phase shifter	Ch. 2	42-45	8, 9, 10	CO1	L2
3	Properties of the Hilbert transform and why they matter	Ch. 2	42-45	11, 12, 13	CO1	L3
4	The pre-envelope, or analytic signal	Ch. 2	45-47	14, 15, 16	CO1	L3
5	The complex envelope: removing the carrier	Ch. 2	47-49	17, 18, 19	CO1	L3
6	Canonical representation: in-phase and quadrature components	Ch. 2	49-52	20, 21, 22	CO1	L3
7	Polar form: natural envelope and phase	Ch. 2	49-52	23, 24, 25	CO1	L3
8	The AWGN channel model and what it assumes	Ch. 7	323-324	26, 27, 28	CO1	L2
9	Geometric representation: signals as points in a vector space	Ch. 7	324-332	29, 30, 31	CO1	L3
10	The Gram-Schmidt orthogonalization procedure	Ch. 7	324-332	32, 33, 34	CO1	L4
11	Energy, distance and the constellation diagram	Ch. 7	324-332	35, 36, 37	CO1	L4
12	Converting the continuous AWGN channel into a vector channel	Ch. 7	332-337	38, 39, 40	CO1	L4
13	The theorem of irrelevance and sufficient statistics	Ch. 7	332-337	41, 42, 43	CO1	L4
14	Maximum likelihood decoding and decision regions	Ch. 7	337-344	44, 45, 46	CO1	L4
15	The correlation receiver	Ch. 7	337-344	47, 48, 49	CO1	L3
16	The matched filter receiver and its equivalence to the correlator	Ch. 7	337-344	50, 51, 52	CO1	L4

Verbatim syllabus lines

- Bandpass Signals to Equivalent Lowpass: Hilbert Transform, Pre-envelopes, Complex envelopes of Band-pass Signals, Canonical Representation of Bandpass signals
- Signalling over AWGN Channels- Introduction, Geometric representation of signals, Gram-Schmidt Orthogonalization procedure
- Conversion of the continuous AWGN channel into a vector channel
- Optimum receivers using coherent detection: ML Decoding, Correlation receiver, matched filter receiver
- Text book 1: Ch 2.8-2.11, Ch 7.1-7.4

- Number of Hours: 10

Provenance

Built from TB1 Sections 2.8-2.11 and 7.1-7.4, the sections the syllabus names. The organising idea is a single change of representation: a bandpass waveform carries no information in its carrier, so the carrier is removed analytically to leave a complex lowpass envelope, and that envelope is then projected onto an orthonormal basis so that a continuous waveform becomes a point in a finite-dimensional space. Once signals are points and noise is a Gaussian cloud, the optimum receiver is forced rather than invented: it is nearest-neighbour decision, and the correlation receiver and matched filter are two circuits computing the same projection.

Module 2 — Digital Modulation Techniques

Driving question: Given a fixed power budget and a fixed band, how many bits per second can I push through, and what does each extra bit cost in reliability?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	The Q function and the shape of every error probability	Ch. 7	344-352	5, 6, 7	CO2	L2
2	BPSK: constellation, generation and bandwidth	Ch. 7	352-357	8, 9, 10	CO2	L2
3	BPSK: coherent detection and bit error probability	Ch. 7	352-357	11, 12, 13	CO2	L4
4	QPSK: constellation and Gray mapping	Ch. 7	357-364	14, 15, 16	CO2	L3
5	QPSK: generation, detection and error probability	Ch. 7	357-364	17, 18, 19	CO2	L4
6	M-ary PSK: constellation and decision regions	Ch. 7	364-370	20, 21, 22	CO2	L4
7	M-ary PSK: error probability and the power-bandwidth trade-off	Ch. 7	364-370	23, 24, 25	CO2	L5
8	M-ary QAM: using the interior of the plane	Ch. 7	370-375	26, 27, 28	CO2	L4
9	M-ary QAM: error probability and comparison with PSK	Ch. 7	370-375	29, 30, 31	CO2	L5
10	BFSK: orthogonal signalling in two dimensions	Ch. 7	375-382	32, 33, 34	CO2	L3
11	BFSK: coherent detection and error probability	Ch. 7	375-382	35, 36, 37	CO2	L4

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
12	Comparing BPSK, QPSK and BFSK on one chart	Ch. 7	415-418	38, 39, 40	CO2	L5
13	Detection of signals with unknown phase	Ch. 7	400-410	41, 42, 43	CO2	L4
14	BFSK using noncoherent detection	Ch. 7	410-411	44, 45, 46	CO2	L4
15	Differential phase-shift keying	Ch. 7	411-415	47, 48, 49	CO2	L4
16	Choosing a modulation scheme for a given link	Ch. 7	415-418	50, 51, 52	CO2	L5

Verbatim syllabus lines

- Digital Modulation Techniques: Phase shift Keying techniques using coherent detection: generation, detection and error probabilities of BPSK and QPSK
- M-ary PSK, M-ary QAM
- Frequency shift keying techniques using Coherent detection: BFSK generation, detection and error probability
- BFSK using Noncoherent Detection, Differential Phase Shift Keying
- Text book 1: Ch 7.6, 7.7, 7.8, 7.12, 7.13
- Number of Hours: 10

Provenance

Built from TB1 Sections 7.5-7.8 and 7.10-7.14, which the syllabus names. Every scheme in this module is the same three questions asked again: where are the constellation points, what is the minimum distance in terms of energy per bit, and what does the receiver need to know about the carrier. The unifying result is that error probability always takes the form Q of a distance divided by the noise spread, so once a constellation is drawn its performance follows without further integration. Noncoherent detection is introduced at the end as the price paid when phase is unavailable, which sets up the trade-off between receiver simplicity and about 1 dB of energy.

Module 3 — Information Theory

Driving question: What is the least I can pay to represent a source, and the most I can push through a channel, before the mathematics says no?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Information as surprise: the logarithmic measure	Ch. 5	207-214	5, 6, 7	CO3	L2

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
2	Entropy: the average information per symbol	Ch. 5	207-214	8, 9, 10	CO3	L3
3	Properties of entropy and the extension of a source	Ch. 5	207-214	11, 12, 13	CO3	L4
4	The source coding theorem	Ch. 5	214-215	14, 15, 16	CO3	L3
5	Prefix codes and the Kraft inequality	Ch. 5	215-223	17, 18, 19	CO3	L4
6	Huffman coding	Ch. 5	215-223	20, 21, 22	CO3	L4
7	Lempel-Ziv coding and the case for dictionary methods	Ch. 5	215-223	23, 24, 25	CO3	L4
8	Code efficiency, redundancy and where compression stops	Ch. 5	215-223	26, 27, 28	CO3	L4
9	Discrete memoryless channels and the transition matrix	Ch. 5	223-226	29, 30, 31	CO3	L3
10	The binary symmetric and binary erasure channels	Ch. 5	223-226	32, 33, 34	CO3	L4
11	Mutual information: what the output tells you about the input	Ch. 5	226-230	35, 36, 37	CO3	L4
12	Properties of mutual information	Ch. 5	226-230	38, 39, 40	CO3	L4
13	Channel capacity	Ch. 5	230-232	41, 42, 43	CO3	L4
14	The channel coding theorem	Ch. 5	232-237	44, 45, 46	CO3	L4
15	The Information Capacity Law	Ch. 5	240-244	47, 48, 49	CO3	L4
16	The bandwidth-power trade-off and the Shannon limit	Ch. 5	240-244	50, 51, 52	CO3	L5

Verbatim syllabus lines

- Information theory: Introduction, Entropy, Source Coding Theorem
- Lossless Data Compression Algorithms, Discrete Memoryless Channels
- Mutual Information, Channel capacity, Channel Coding Theorem
- Information Capacity Law (Statement)
- Text book 1: Ch 5.1-5.10
- Number of Hours: 10

Provenance

Built from TB1 Sections 5.1-5.10, which the syllabus names. The module has two halves that mirror each other. The source-coding half asks how few bits a source needs and answers with entropy, achieved by Huffman and Lempel-Ziv. The channel-coding half asks how many bits a channel can carry and answers with capacity, the maximum of mutual information over input distributions. Both are existence results: they say what is possible without saying how, which is exactly the gap that Modules 4 and 5 fill. The Information Capacity Law at the end unifies them and produces the Shannon limit of -1.6 dB, the number every coding scheme is ultimately measured against.

Module 4 — Error Control Coding: Linear Block and Cyclic Codes

Driving question: How do I add redundancy so that the receiver can find and repair errors it cannot even see?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Forward error correction versus automatic repeat request	Ch. 10	577-579	5, 6, 7	CO4	L3
2	Linear block codes: the definition and why linearity matters	Ch. 10	582-586	8, 9, 10	CO4	L3
3	The generator matrix	Ch. 10	582-586	11, 12, 13	CO5	L3
4	The parity check matrix	Ch. 10	582-586	14, 15, 16	CO5	L4
5	Systematic form and why encoders prefer it	Ch. 10	582-586	17, 18, 19	CO5	L3
6	The syndrome and its properties	Ch. 10	586-590	20, 21, 22	CO4	L4
7	Minimum distance and minimum weight	Ch. 10	586-590	23, 24, 25	CO4	L4
8	Error detection and correction capability	Ch. 10	586-590	26, 27, 28	CO4	L4
9	Syndrome decoding and the standard array	Ch. 10	590-593	29, 30, 31	CO5	L4
10	Hamming codes	Ch. 10	590-593	32, 33, 34	CO5	L4
11	Cyclic codes: definition and properties	Ch. 10	593-597	35, 36, 37	CO4	L4
12	The generator polynomial	Ch. 10	593-597	38, 39, 40	CO5	L4

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
13	Parity check polynomial and matrix forms	Ch. 10	597-601	41, 42, 43	CO5	L4
14	Systematic cyclic encoding with a shift register	Ch. 10	597-601	44, 45, 46	CO5	L5
15	Syndrome computation for cyclic codes and a worked example	Ch. 10	601-605	47, 48, 49	CO5	L5
16	Cyclic redundancy checks and burst error detection	Ch. 10	601-605	50, 51, 52	CO4	L5

Verbatim syllabus lines

- Error Control Coding: Error Control Using Forward error Correction
- Linear Block Codes: Definitions, Matrix Descriptions, Syndrome and its properties
- Minimum distance Considerations, Syndrome Decoding, Hamming Codes
- Cyclic Codes: Properties, Generator and Parity Check Polynomial and matrices, Encoding, Syndrome computation, Examples
- Text book 1: Ch 10.2, 10.4, 10.5
- Number of Hours: 10

Provenance

Built from TB1 Sections 10.1-10.5, which the syllabus names. The module is the constructive answer to Module 3: capacity said good codes exist, and this is the first family that can actually be built and decoded. The organising idea is linear algebra over $GF(2)$ — codewords form a subspace, the generator matrix is its basis, the parity check matrix annihilates it, and the syndrome is the image of the received word under that annihilator. Everything about detection and correction capability follows from one number, the minimum distance. Cyclic codes then add structure so that matrix multiplication becomes polynomial arithmetic and the encoder collapses to a shift register.

Module 5 — Convolutional Codes

Driving question: How do I code a continuous stream with no block boundaries, and how does the decoder find the best path without checking every one?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	The convolutional encoder: memory instead of blocks	Ch. 10	605-608	5, 6, 7	CO4	L2
2	Constraint length, rate and generator sequences	Ch. 10	605-608	8, 9, 10	CO4	L3
3	Impulse response and generator polynomials	Ch. 10	605-608	11, 12, 13	CO5	L4

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
4	Time-domain encoding: convolution by hand	Ch. 10	605-608	14, 15, 16	CO5	L4
5	The code tree and why it does not scale	Ch. 10	608-613	17, 18, 19	CO4	L3
6	The trellis diagram	Ch. 10	608-613	20, 21, 22	CO5	L4
7	The state diagram and the transfer function	Ch. 10	608-613	23, 24, 25	CO5	L5
8	Recursive systematic convolutional codes	Ch. 10	610-613	26, 27, 28	CO4	L5
9	Free distance and coding gain	Ch. 10	610-613	29, 30, 31	CO4	L5
10	Optimum decoding: what the decoder is actually trying to do	Ch. 10	613-614	32, 33, 34	CO4	L4
11	The Viterbi algorithm: add, compare, select	Ch. 10	614-618	35, 36, 37	CO5	L4
12	A full Viterbi decoding example	Ch. 10	618-623	38, 39, 40	CO5	L5
13	Traceback, truncation depth and decoding delay	Ch. 10	618-623	41, 42, 43	CO5	L5
14	Hard versus soft decision decoding	Ch. 10	618-623	44, 45, 46	CO5	L5
15	Choosing and deploying a convolutional code	Ch. 10	618-623	47, 48, 49	CO5	L5
16	Block codes and convolutional codes compared	Ch. 10	605-623	50, 51, 52	CO4	L5

Verbatim syllabus lines

- Convolutional Codes: Convolutional Encoder, Code tree, Trellis Graph and State graph
- Recursive systematic Convolutional codes
- Optimum decoding of Convolutional codes
- Maximum Likelihood Decoding of Convolutional codes: The Viterbi Algorithm, Examples
- Text book 1: Ch 10.6, 10.7, 10.8
- Number of Hours: 10

Provenance

Built from TB1 Sections 10.6-10.8, which the syllabus names. The shift from Module 4 is from blocks to streams: a convolutional encoder has memory, so each output depends on several past inputs and there is no block boundary to align to. The same encoder admits three equivalent pictures — tree, trellis and state diagram — and the module spends time on why the trellis is the useful one: it is the only one that does

not grow with sequence length, which is precisely what makes the Viterbi algorithm linear rather than exponential. Free distance replaces minimum distance as the figure of merit, and the module closes on hard versus soft decision decoding, where 2 dB is available for the cost of keeping more than one bit from the demodulator.

Note on textbook page numbers

All TB1 chapter and page references were taken from the supplied Digital Communication Systems PDF's own table of contents; the file extracted to roughly 2.1 million characters of machine-readable text, so no TB1 page reference in this document is from memory.